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A High-precision Edge Detection Algorithm Based on Improved Sobel Operator-Assisted HED

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***Abstract*— In response to the problem of poor detection performance of the traditional Sobel operator in edge detection, a high-precision edge detection algorithm based on Sobel operator-assisted Holistically-nested Edge Detection (HED) network was proposed, which is named ISAHED here. The algorithm is designed as follows. Firstly, an improved Sobel operator based on the traditional Sobel operator was proposed through adding gradient direction. Then, the image output by the improved Sobel operator is taken as the ground-truth image and as the end-to-end comparison object in a HED-like network. Finally, the ISAHED algorithm outputs edge images with better quality through iterative learning of the HED-like neural network. The experimental results prove that the proposed ISAHED algorithm has better edge detection performance by visually observing the final edge images and further evaluating the values of FOM. This work provides an important reference for the further development of the edge detection technology.**

I. Introduction

With the increasing development of edge detection technologies, it has been of important application value in medical image processing, environmental monitoring and other fields [1-3]. Hence, the edge detection technology plays a crucial role. The industry increasingly needs the edge detection methods that can improve the reliability of the edge detection, enhance the edge feature expression, suppress the background interference, and adapt to the complex environments [4-7]. However, the traditional edge detection operators cannot meet the above requirements, and how to find an edge detection method with the high precision, generalization, complete contour and rich details is one of the research hotspots in the field of the computer vision in recent years.

The current mainstream edge detection methods include gradient-based methods, model-based methods and deep learning-based methods [8-10]. The first two methods are usually called as traditional edge detection methods, and the traditional edge detection methods have many limitations due to their simple principles, such as being too sensitive to noise, inaccurate edge detection, et al. The deep learning method has gradually become the mainstream of edge detection methods due to its powerful capabilities. With the development of deep learning technologies, the deep learning is widely used in the edge detection. The improvement of traditional operators and the combination of traditional operators and deep learning have become new hotspots in edge detection. For instance, ShuKang Chen, et al. [11] proposed an enhanced adaptive Sobel edge detector based on the improved genetic algorithm and non-maximum suppression, and the results show that the enhanced adaptive Sobel edge detector outperforms other comparative algorithms in terms of accuracy, denoising power and time cost. Chenxing xue, et al. [12] obtained the trained edge detection operator through the convolutional neural network (CNN), and confirmed that the edge detection operator based on the CNN has a higher edge detection effect than the classic operator. Harshith V, et al. [13] proposed an algorithm based on the Canny operator to be used in CNN networks. Through the algorithm, the edges are strengthened, which greatly improves the accuracy of the facial expression recognition in experiments. In sum, the above three studies are about the improvement of operators, the use of CNN to generate better operators, and the combination of operators and deep learning. Their edge extraction effects are sequentially better. Therefore, it is necessary to combine deep learning technologies with Sobel operator for edge detection algorithms.

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A new edge detection algorithm based on the Sobel operator-assisted HED network (ISAHED) is proposed in this work. Comparative analysis confirms that the proposed ISAHED algorithm has higher edge detection performance, which refers to the output images having more complete contours and richer details.

II. Related Works

The Sobel operator is the most classic first-order differential convolution operator. The principle of the traditional Sobel operator is as follows, the images are weighted and then subtracted through convolution kernels in both horizontal and vertical directions. As shown in Fig.1, the traditional Sobel operator is a convolution kernel of the 3×3 template. It takes top, bottom, left and right fields into consideration, the image is processed in the horizontal and vertical directions shown as in Fig.1a and Fig.1b, and the gradient value is obtained by the weighted difference, only the gradient value is considered in the processed pixels.

-1	-2	-1
0	0	0
1	2	1

(a) Horizontal Direction

-1	0	1
-2	0	2
-1	0	1

(b) Vertical Direction

Fig.1 Convolution kernels of the traditional Sobel operator in (a) horizontal and (b) vertical directions.

G1	G2	G3
G4	G5	G6
G7	G8	G9

Fig.2 Gradient value templates of 3×3 for the traditional Sobel operator.

The image is divided into multiple gradient value templates of 3×3, each pixel is respectively marked as G1 to G9 as shown in Fig.2. The image is processed by the convolution kernel of the Sobel operator. The specific process is represented in Equation (1).

$$\begin{cases} \Delta_x f(x,y) = [G7+2G8+G9]-[G1+2G2+G3] \\ \Delta_y f(x,y) = [G3+2G6+G9]-[G1+2G4+G7] \end{cases} \tag{1}$$

The gradient value of the $G[f(x,y)]$ pixel, namely the gradient value of G5, can be expressed as the root of the sum of the square of the gradient values processed by each convolution kernel, as shown in Equation (2).

$$G[f(x,y)] = \sqrt{\Delta_x f^2(x,y) + \Delta_y f^2(x,y)} \tag{2}$$

III. ISAHED Edge Detection Algorithm

A. Improved Sobel operator

In this work, the Sobel operator is improved as follows. First, the images are processed by Gaussian filtering to reduce the influence of noise. In the 3×3 template, each of the pixel obviously has four gradient directions, so the gradient value template of 3×3 was positioned in 0 degree, 45 degree, 90 degree and 135 degree, as shown in Fig.3. Then, on the basis of the vertical and horizontal gradient values processed by the traditional Sobel operator (namely, 0 degree and 90 degree), the convolution kernels of two additional optimized operators[14] (namely, 45 degree and 135 degree) are added to perform the image convolution processing, and the convolution kernel in two directions of 45 degree and 135 degree is shown as Fig. 4a and Fig. 4b. The gradient values of the four directions are shown in Equation (3). Based on the four-directional gradient value expressed by Equation (3), this work proposes that the gradient value of the pixel point can further be expressed by Equation (4). In Equations (3) and (4), $\Delta_{0^\circ} f(x,y)$, $\Delta_{45^\circ} f(x,y)$, $\Delta_{90^\circ} f(x,y)$, $\Delta_{135^\circ} f(x,y)$ represent the gradient values of the image after being processed by four convolutional kernels, respectively.

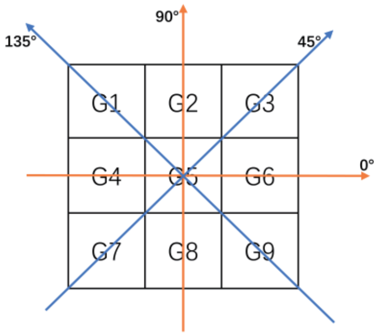

Fig.3 The gradient direction template of the improved Sobel operator.

-2	-1	0
-1	0	1
0	1	2

(a) 45°

0	1	2
-1	0	1
-2	-1	0

(b) 135°

Fig.4 The convolution kernel in two directions of (a) 45 degree and (b) 135 degree.

$$\begin{cases} \Delta_{0^\circ} f(x,y) = [G7+2G8+G9]-[G1+2G2+G3] \\ \Delta_{45^\circ} f(x,y) = [G6+2G9+G8]-[G2+2G1+G4] \\ \Delta_{90^\circ} f(x,y) = [G3+2G6+G9]-[G1+2G4+G7] \\ \Delta_{135^\circ} f(x,y) = [G2+2G3+G6]-[G4+2G7+G8] \end{cases} \tag{3}$$

$$\mathrm{G}[f(x,y)] = \sqrt{\Delta_{0^\circ} f^2 + \Delta_{45^\circ} f^2 + \Delta_{90^\circ} f^2 + \Delta_{135^\circ} f^2} \tag{4}$$

After that, the processed edge map is binarized by applying a threshold. Each pixel in the image is classified into one of two categories, such the edge map of the image is obtained. The specific operation process is shown in Equation (5). In Equation (5), the edges represent the edge detection image processed by Equation (4), and threshold is expressed as the limit value. It is evident that these pixels are classified as edge points '1' or non-edge points '0' by comparing the gradient value of each pixel with the threshold.

At this point, compared with the processing results of the traditional Sobel operator, an edge image with better multi-directional edge detection performance is obtained through the improved Sobel operator.

$$\begin{cases} edges = 1 & (edges > threshold) \\ edges = 0 & (others) \end{cases} \tag{5}$$

B. The architecture of a HED-like neural networks

A HED-like neural network is used to achieve the training of the end-to-end and multi-scale learning in this work. Based on the VGG-16 network structure [15], a simplified HED-like algorithm model was set up with principle of HED network, which is used for edge detection tasks as shown in Fig.5. The network consists of five convolutional blocks, which are set up to extract multi-scale and more edge features. Each block comprises multiple convolutional layers and pooling layers to increase the model's receptive field.

The first convolutional block consists of four convolutional layers and one pooling layer. Each convolutional layer includes 64 convolutional kernels of 1×1 magnitude to extract features from the original images.

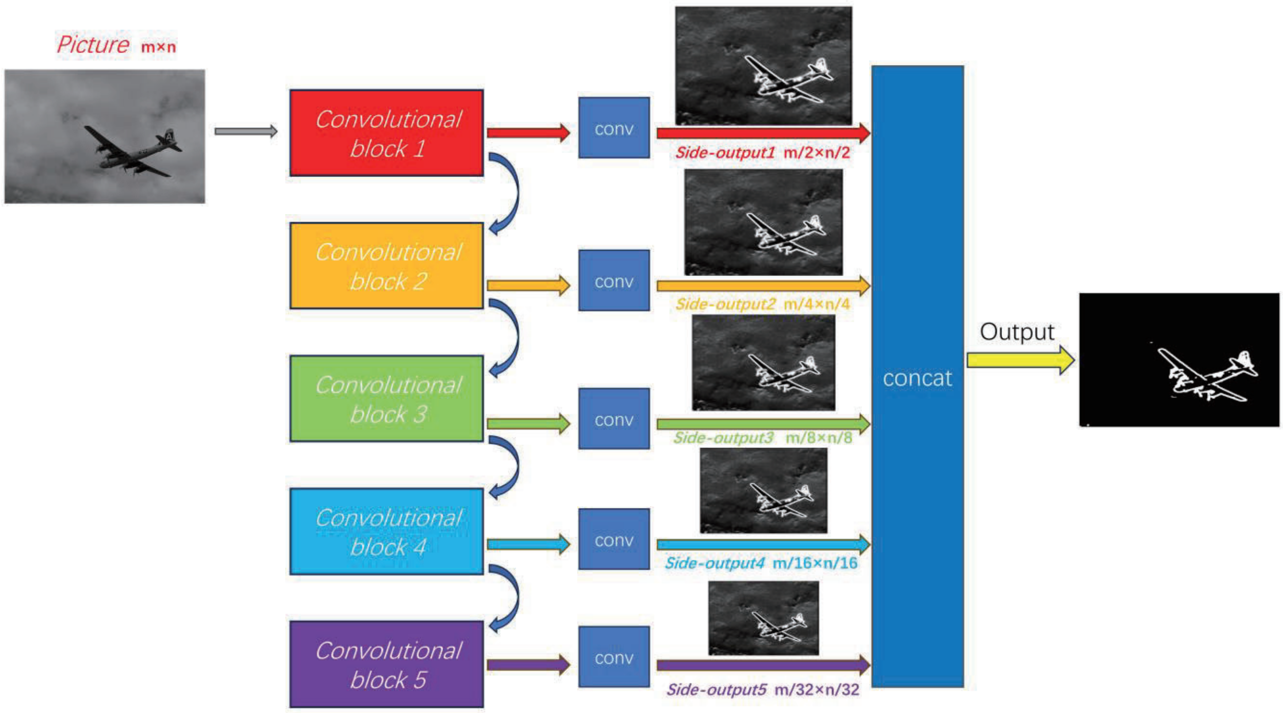

Fig.5 The architecture of the HED-like neural network

The principles of other convolutional blocks are similar to that of the first convolutional block, but the number of convolutional kernels and the number of convolutional layers are different from those of the first convolutional block. Then through additional convolutional layers, the edge features processed by the five convolutional blocks of the network are output as Side-outputs at five different scales. These outputs are upsampled. Score1-5 respectively represent the feature maps obtained by upsampling five different side-outputs, and the Score1-5 are converted into edge detection images of the same size as the original image by using bilinear interpolation method shown in Equation (6). Where $f_{horz}(x,y)$ is processed in the horizontal direction, $f_{vert}(x,y)$ is processed in the vertical direction, and (x, y) is the target interpolation points. f_{11}、f_{12}、f_{21} and f_{22} are the four neighboring pixels of the target interpolation point (x, y) with coordinates (x1, y1), (x1, y2), (x2, y1), (x2, y2). Score1-5 are fused through Concatenation operation, each of them are weighted and combined into an edge detection image for comparison with the ground truth. The processing of Concatenation fusion is shown in Equation (7). The C here indicates concatenation along the same channel dimension of images, forming a new tensor as the final edge detection result.

$$\begin{cases} f_{horz}(x,y) = \dfrac{(x_2-x_1)(x_2-x)\cdot f_{11} + (x-x_1)\cdot f_{21}}{(x_2-x_1)} \\ f_{vert}(x,y) = \dfrac{(y_2-y_1)(y_2-y)\cdot f_{horz}(x,y_1) + (y-y_1)\cdot f_{horz}(x,y_2)}{(y_2-y_1)} \\ f(x,y) = f_{vert}(x,y) \end{cases} \tag{6}$$

$$C = [Score1, Score2, Score3, Score4, Score5] \tag{7}$$

The accuracy of each iteration's prediction is measured by BCE Loss (Binary Cross Entropy Loss), as shown in Equation (8). The *pt* indicates the model prediction value, *target* represents the label value, and ω is the weight value which is usually set into 1, and it was also set to 1 in this experiment. After calculating the loss value, the Adam optimizer is used to backpropagate and adjust the model parameters to minimize the loss function, thereby achieving the effects of end-to-end training and multi-scale learning of the entire network.

$$\begin{cases} L(pt, target) = -\omega \times (target \times \ln(pt)) + (1-target) \times \ln(1-pt) \\ loss = \dfrac{1}{N}\sum L \end{cases} \tag{8}$$

C. Implementation process

BSDS500 (Berkeley Segmentation Data Set 500) is a standard benchmark for contour detection and a public image database that includes the original image and the manually labeled edge images of the original image. In the ISAHED algorithm, firstly, combined with the advantages of the Sobel operator, the edge images obtained by using the improved Sobel operator in this work to perform the edge detection on all the images in BSDS500 are now used as ground-truth in BSDS500. Then, the images processed by each convolution and pooling operation of the network are compared to achieve learning from multi-scale images, after multiple iterations, the best network is obtained. Subsequently, the predicted image is put into the network as the input, and the network directly performs a predicted operation on the pixels of the image and outputs the grayscale image. At the end of the work, the grayscale image is thresholded by the network to obtain the final edge detection result for visualization.

IV. Simulation Result and Analysis

The model loss of the proposed algorithm is shown as Fig.6. In the model training, from 1 to 15 times, the training loss and validation loss continue to decrease with the increasing number of iterations, so the model need to continue training, and from 15 to 25 iterations, the training loss and validation loss tend to level off. To prevent overfitting, the intermediate value of 20 is taken as the number of iterations for this experiment, and the model already has good performance.

As shown in Fig.7, it shows the different effects of different edge detectors on detecting the edges of images. Using five different images under the same threshold, the following visual edge detection effects of different edge detectors are obtained, where the columns represent different images and the rows represent different detection methods. It can be clearly observed that compared with other edge detectors, the proposed algorithm has richer edge detection information and a more complete edge detection profile.

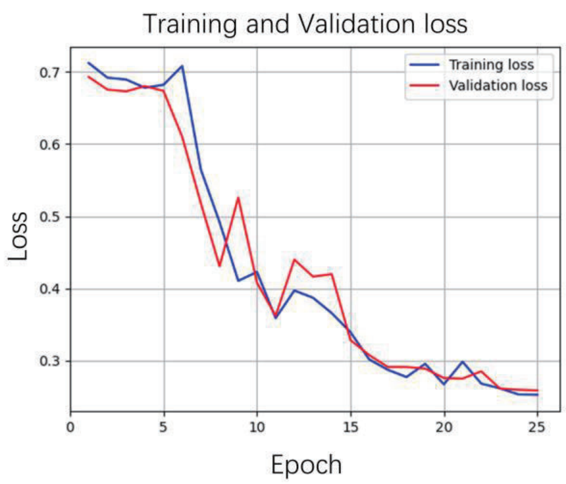

Fig.6 Model loss of the proposed ISAHED algorithm.

The five images are randomly selected from the BSDS500 database. Fig.7 shows the edge detection effect of the five images performed through various edge detectors. From a visual point of view, in the first image, due to sunlight, the objects on the shaded side are generally darker in brightness, and very little differences in the grayscale of adjacent objects in the image was found. For example, the edge between the window on the left and the wall, or the edge between the middle railing and the door. Such edges are difficult to identify for gradient-based operators. Under the same threshold, it is not difficult to observe that the edge contour is missing in the traditional method, especially on the Sobel operator and the Perwitt operator, the detection effect is very poor. For the improved Sobel operator, the edge is relatively complete, but there are still breakpoints and some edge features are missing. And for the algorithm proposed in this work, not only the outlines are complete, but there are more edge features, such as the door in the first image and the upstairs windows. The detection effects of each operator from the first image to the fourth image in Fig.7 are similar. In the edge detection of the fifth image, it is quite evident that under the same threshold, the detection effect of the traditional edge detection algorithm is unsatisfactory and the contours are missing seriously, even after improving the Sobel operator for edge detection, significant parts of the edges including outer contours are still being lost, let alone delve into internal details. while the edge detection effect of the algorithm proposed in this work still has complete contours and rich details.

The Figure of merit (FOM) value is one of the metrics used to quantitatively compare the processing performance of edge detectors in the field of image processing. The specific algorithm is shown in Equation (9). Where Gt is the number of edge pixels in the real edge image (namely, ground truth), Dt represents the number of edge pixels in the detected edge image, and TP is the true positives, which is the number of edge pixels that are correctly detected. FP is the false positives, which is the number of edge pixels that are incorrectly detected. $dGt(p)$ is the distance from pixel p to the nearest true edge pixel. k is a constant to be usually set into a value of 0.1, and it was also set into 0.1 in this work. Thus, the higher the FOM value (that is, close to 1), the better the edge detector performance.

$$FOM = \frac{1}{\max(|Gt|,|Dt|)} \times (TP + \sum_{P \in FP} \frac{1}{1 + k \times [dGt(p)]^2}) \tag{9}$$

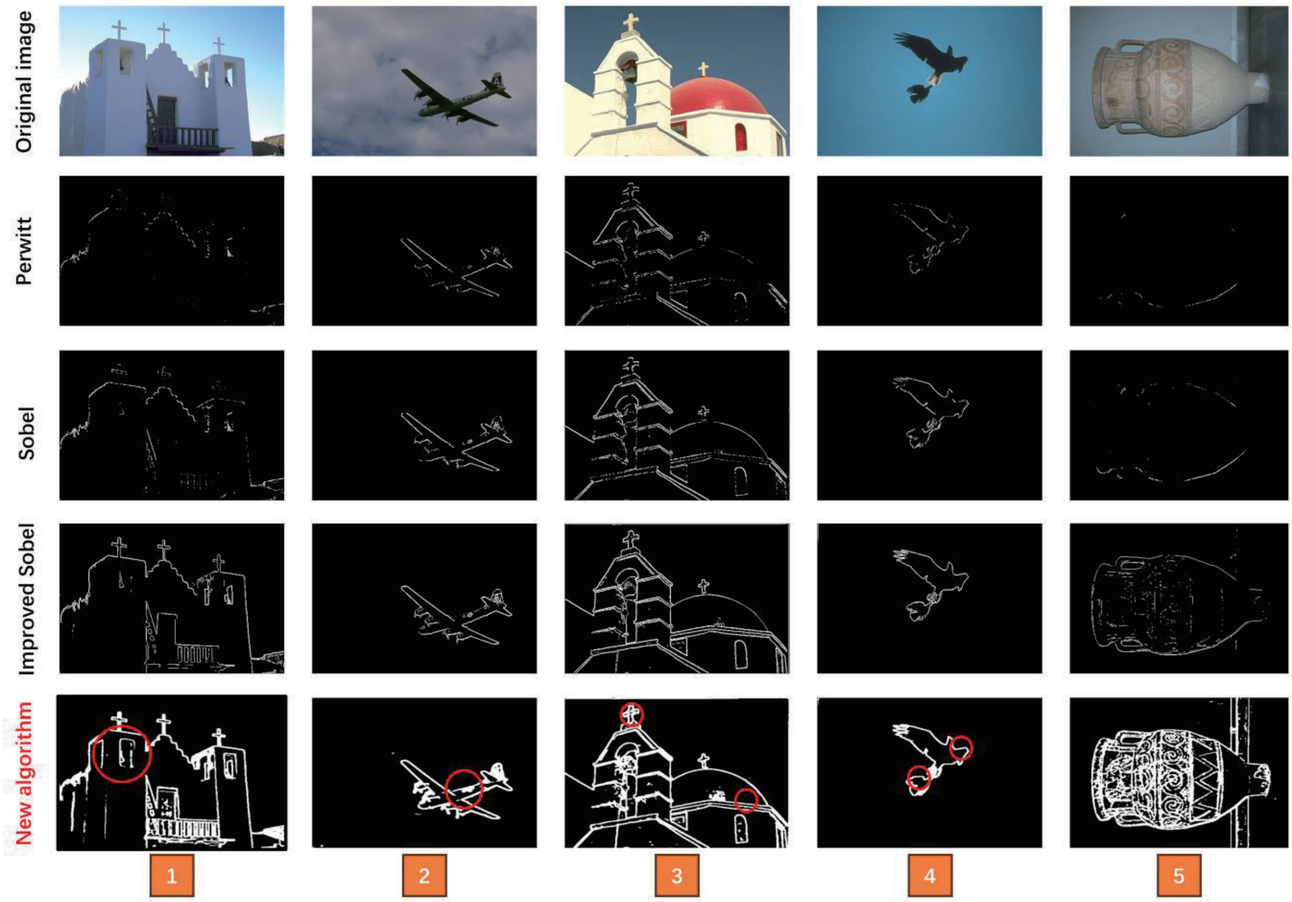

Fig.7 Comparative analysis of edge detection results between the ISAHED algorithm and other algorithms.

In this work, the FOM index is used to quantitatively analyze the edge detection effect of each edge detector. The performance value of each edge detector in five different groups of images is calculated, as shown in Fig.8. The numbers 1-5 in this figure correspond to the original figures 1-5 in Fig. 7. It is obvious that the FOM value of the edge detector for the new algorithm can reach up to 9 times that of the improved Sobel operator and it is far better than two traditional operators. Therefore, it has been quantitatively confirmed that the proposed algorithm has the better edge detection effect.

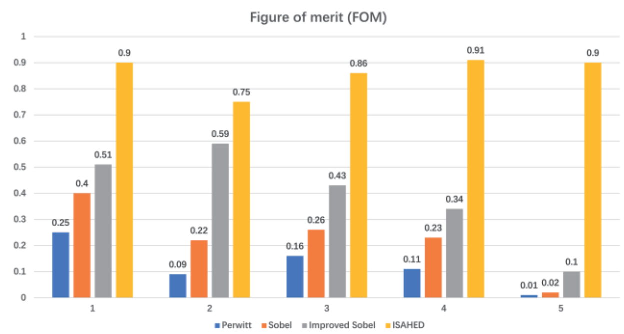

Fig.8 Comparative analysis of FOM values between the ISAHED algorithm and other algorithms.

In short, the model has achieved good performance through 20 iterations. Furthermore, combined with Fig.7 and Fig.8, it is evident that the proposed algorithm has the richer edge detection information and the more complete edge detection profile compared with other edge detectors.

V. Conclusion

In this work, a new method of the edge detection based on improved Sobel operator-assisted neural network is proposed. In the method, the images detected by the improved Sobel operator are taken as the ground-truth images and as the end-to-end comparison objects in HED-like networks. Judging from the experimental results, the proposed algorithm can obtain a more complete edge in complex situations and can detect richer edge detection details. The experimental results fully confirm the advantages of the algorithm proposed in this work compared to other traditional algorithms in the field of image edge detection. This work provides meaningful reference for high-performance edge detection and opens a new avenue in image processing.

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